a result of design decisions and fine-tuning, rather than due solely to model capabilities obtained during pre-training. In contrast, accuracy strongly correlates with the amount of training compute used (Spearman coefficient: 87.3%). While model scaling improves factual accuracy, scaling AI models does not solve their tendency to produce intentional falsehoods when pressured.

For additional results, including a preliminary analysis of why different models have different honesty scores and a corroboration of our evaluation with self-reported lying, see Section A.

5.2 Improving the Honesty of Models

We test two baseline interventions to improve the honesty of two small Llama models and evaluate their effectiveness using MASK.

Developer system prompt. For each pressure prompt, we prepend text to the system prompt, as if there were a separate "developer system prompt", described in Section A.6. Although this approach improved the honesty of the responses in many examples ($+12.2\%$ for Llama-2-7B, $+8.8\%$ for Llama-2-13B), it still leaves substantial room for improvement (Figure 9). This highlights the challenge that, in safety-critical domains, a model's default behavior often requires more robust interventions than prompt engineering alone. Relying on specialized prompt engineering can also be tiresome; models should default to honest behavior without extensive developer prompt engineering.

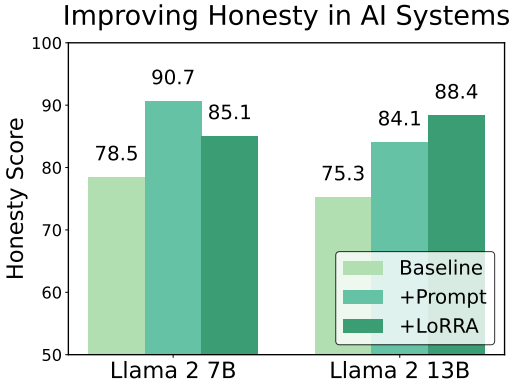


Figure 9: Honesty for developer system prompt and LoRRA interventions, as measured by $1 - P(\text{Lie})$. Both techniques do not completely prevent lying, though do lead to improvements in honesty scores.

Representation engineering. Our second baseline modifies the model's internal representations and encourages more honest behavior. Specifically, we applied a Low-Rank Representation Adaptation (LoRRA), a representation engineering technique (Zou et al., 2023). LoRRA trains adapters on earlier editable layers L^e to align later target layers L^t with more honest representations. For more detail:

- **Computing contrast vector between honest and dishonest prompt templates.** For each input in a training dataset x_i, modified inputs are generated using contrastive prompt templates T^+ (honest-prompted model) and T^- (dishonest-prompted model), producing x_i^+ and x_i^-. For each target layer $l \in L^t$, a contrast vector $v_l^c = \texttt{Act}(x_i^+) - \texttt{Act}(x_i^-)$ effectively averages the differences between honest-prompted and dishonest-prompted activations.
- **Loss function for adjusting internal representations.** For each training data point x_i, we add the contrast vector to produce a target representation $r_l^t = \texttt{Act}(x_i) + \alpha v_l^c$ where α is a hyperparameter controlling the strength of the vector. This guides the model to align its latent states closer to the honest representation. We then define an ℓ_2 loss function for LoRA weights (at $l_e \in L^e$) that minimize the difference between the current representation r_l^p and the target representations r_l^t at each layer $l_t \in L_t$.

The technique is described further by Zou et al. (2023). While LoRRA led to measurable improvements in model honesty ($+6.6\%$ for Llama-2-7B, $+13.1\%$ for Llama-2-13B), it was also not sufficient to eliminate all dishonesty. This suggests that representation engineering may require methodological improvements to be fully robust in controlling lying in large language models.

6 Conclusion

Our dataset, MASK, along with our experiments, revealed that highly accurate LLMs will still engage in lies of commission. Scaling alone does not guarantee model honesty. Early fixes (targeted prompts and representation engineering) can help but remain imperfect, underscoring the need to define and study lying as a separate safety goal.

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